

KEYSTRA ENERGY GROUP

SOLUTION 3:

SUPPLY CHAIN RISK AND RESILIENCE

Predictive Early Warning and Risk Management System

BUSINESS QUESTION

Which suppliers present the greatest risk to Keystra's operations, how could their failure propagate across critical materials, assets and business units, what financial exposure could be affected, and where should management intervene first?

Areas of Investigation>>

- Procurement & Spend Risk

Outstanding purchase orders → procurement exposure → committed order value → priority and status of obligations.

- Supplier Performance & Failure Risk

Supplier reliability → delivery behaviour → historical performance patterns → predicted supplier failure probability.

- Materials & Inventory Risk

Material criticality → supplier dependency → inventory exposure → potential material availability consequences.

- Logistics & Fulfilment Risk

Delivery performance → outstanding quantities → delay and fulfilment behaviour → potential supply disruption.

- Operational & Business Impact

Critical asset exposure → business-unit exposure → downstream risk propagation → financial exposure → potential operational consequences of supplier disruption.

KEY ANALYTICS AREA

- Descriptive & Diagnostic Assessment
- Supplier Failure Prediction
- Risk Propagation & Business Impact

- **Financial Exposure Assessment**
- **Integrated Risk Scoring & Prioritisation**
- **Risk Driver Explanation**
- **Management Action & Intervention**

1. Introduction

2. Business Understanding

- Business context
- Business Objectives and value

3. Data Understanding and Preparation

- Data requirements and sources
- Data preparation and engineering

4. Risk Analytics and Predictive Modelling

- Establishing the supplier failure outcome
- Staged supplier risk framework
- Turning supplier risk signals into a predictive dataset
- Turning supplier risk signals into a predictive early warning model
- Turning supplier failure predictions into actionable risk warnings

5. From Supplier Risk to Early Warning

- Understanding why the risk matters
- Tracing supplier risk through the supply chain
- From risk assessment to intervention
- Turning risk into an early warning
- What keystra can now do

6. Supplier Risk Register

- Highest risk supplier
- High early warning group
- Emerging risk: suppliers keystra should not ignore
- The watchlist population
- Low current risk

The previous solution Pipeline Integrity & Operational Risk Intelligence, established an asset level view of operational risk by examining how adverse conditions within critical infrastructure could develop into failures and operational consequences. That analysis was developed using a publicly available pipeline dataset from the United States. The dataset reflects U.S. pipeline infrastructure, regulatory, and operating conditions and therefore does not represent Keystra's actual physical asset footprint in Nigeria. It was intentionally used as an external data source to demonstrate how risk analytics methodologies can be applied to energy infrastructure data. The underlying risk relationships, analytical techniques, and decision logic were treated as transferable rather than being presented as a direct representation of Keystra's operations.

The analysis therefore established one side of the risk equation: what happens when risk develops within the asset. The next step is to examine the dependencies that exist before the risk reaches the asset.

In an energy business, the ability of a critical asset to operate reliably is partly dependent on conditions outside the asset itself. Suppliers provide the equipment and materials required to maintain operations; procurement creates financial commitments; logistics determines whether those requirements arrive when needed; and inventory determines whether the organisation can absorb delays or disruption. This creates a different type of risk: a supplier can become a threat to operational continuity before an asset itself shows signs of failure.

The difficulty is that supplier risk is not determined by supplier performance alone. A moderately unreliable supplier supporting a non-critical material may have limited consequences, while the failure of another supplier could simultaneously affect critical materials, multiple assets, several business units, and substantial financial commitments. Supply Chain Risk & Operational Resilience was therefore developed as a risk analytics framework for identifying these high consequence dependencies and determining where intervention is most needed.

The analysis combines supplier behaviour, predictive failure probability, procurement exposure, material criticality, asset dependency, business-unit reach, downstream propagation, and financial exposure into a single risk assessment. This allows the analysis to move beyond identifying who is risky to determining why the risk matters, what could be affected if the supplier fails, and how significant the resulting exposure could be.

The final analytical stage brings these dimensions together into an early warning and management action engine. Suppliers are ranked according to their overall risk significance, their dominant risk drivers are identified, potential downstream consequences are captured, and each prioritised risk is translated into an appropriate level of management urgency and intervention.

BUSINESS CONTEXT

Keystra's operations rely on suppliers to provide the materials, equipment, and services needed to keep its assets and business activities running. When a supplier begins to experience problems, the impact may go beyond the supplier itself affecting material availability, asset operations, business activities, and financial commitments.

However, supplier risk cannot be understood from performance indicators alone. The significance of a supplier depends on the role it plays within the wider organisation and the consequences that could arise if its supply becomes unreliable. A supplier supporting a non-critical requirement may have limited business significance, while another supporting critical materials, multiple assets, or several business units may represent a substantially greater exposure even when their performance indicators appear similar.

The business challenge was therefore to distinguish supplier risk from material supplier risk, identifying where deterioration could translate into meaningful operational or financial consequences.

BUSINESS OBJECTIVES AND VALUE

The analysis was designed around three decision questions:

- Which suppliers are most likely to become a source of disruption?
Assess supplier behaviour and predicted failure probability to identify emerging supplier risk.
- What would be exposed if they failed?
Trace supplier dependencies through materials, assets, business units, downstream operations, and financial commitments.
- Which risks require management attention first?
Combine likelihood and consequence to establish an integrated risk position, explain the factors driving it, and determine appropriate intervention urgency.

These questions formed the basis for the predictive, propagation, exposure, prioritisation, and management-action components of the solution.

The solution provides Keystra with a way to view supplier risk in terms of potential business consequence rather than supplier performance alone. It enables management to identify emerging risks earlier, distinguish material exposures from lower consequence supplier issues, and understand the potential reach of a supplier failure before it develops into a wider operational disruption.

This supports a more targeted risk response: earlier identification, clearer understanding of exposure, and better allocation of management attention and mitigation resources. The broader value is a shift from reactive supplier monitoring to proactive management of supplier driven operational risk.

DATA REQUIREMENTS AND SOURCES

To understand where supplier related risk could affect Keystra, the analysis needed to bring together information from different parts of the supply chain. Looking at supplier performance alone would not be enough, we also needed to know what each supplier provides, what Keystra has committed to purchase, materials are available, and which assets and business units depend on those materials.

The data requirements were therefore built around six main areas: supplier activity, materials, procurement, quality, inventory, and supplier risk outcomes.

Relationships between suppliers and materials, and between materials and assets, were also required so that the potential consequences of supplier disruption could be traced through the business. The Asset and Business Unit structures developed in Solution 1 were retained for this analysis. This meant Solution 4 could build on the existing Keystra data foundation and connect supply chain risks to the assets and business units already defined, rather than creating them again.

The remaining data was engineered specifically for the Solution 4 use case. This included the supplier and material dimensions, failure types, supplier–material and asset–material relationships, purchase order activity, quality events, inventory snapshots, and supplier risk outcomes.

The analysis covers Keystra’s performance from 1 January 2022 to 31 July 2026, providing the historical view needed to understand how supplier conditions changed over time. The same period also supports the development of historical risk indicators and the identification of suppliers whose current behaviour suggests a potential failure within the following 90 days.

The resulting data foundation was designed to answer more than “Which suppliers are risky?” It allowed the analysis to examine why they are risky, what they support, what could be exposed if they fail, and how significant that risk could be to Keystra.

DATA PREPARATION AND ENGINEERING

The data for this solution was engineered and organised into a single analytical structure covering Keystra’s 2022–2026 analysis period. The Asset and Business Unit dimensions already developed in Solution 1 were reused, while the remaining datasets were built around the specific requirements of the supply chain risk analysis.

The data model was built step by step, with each table checked and validated before moving to the next. This made it possible to connect suppliers to the materials they provide, the procurement activity around those materials, inventory and quality conditions, and ultimately the risk outcomes used in the analysis.

DIMENSION TABLES DEVELOPMENT

1. DATE DIMENSION

This was the first table created because time is central to the analysis. It covers every day from 1 January 2022 to 31 July 2026, giving us 1,673 dates across the full analysis period.

The table provides the common time reference for tracking supplier behaviour over time, calculating rolling 30, 90, 180, and 365-day measures, and creating the temporal splits required for the supplier failure model and its 90-day forward looking risk label.

2. ASSET DIMENSION

The existing assets from Solution 1 was retained to maintain consistency across the portfolio. It contains 13 assets mapped to their respective business units:

BUSINESS UNITS	ASSETS
Upstream	Offshore Platform A, Offshore Platform B, Offshore Platform C
Midstream	Pipeline Network Alpha, Pipeline Network Bravo, Gas Compression Station, Gas Processing Plant
Downstream	Refinery Complex, Fuel Distribution Terminal
LNG	LNG Export Terminal
Renewables	Solar Farm North, Solar Farm South, Wind Farm

This structure was carried forward into Solution 4 to establish the link between supplier/material dependencies and the assets potentially affected by supply disruption.

3. SUPPLIERS AND MATERIAL DIMENSIONS

Once the date and asset structure were in place, the next step was to establish the two main entities around which the supply chain analysis would be built: suppliers and materials.

Dim_Supplier contains 90 suppliers and serves as the master reference for supplier related activity throughout the solution. It was designed to hold supplier information separately from the risk and predictive outputs that would be generated later. This

separation was important because the supplier failure model needed to learn from historical behaviour without being given information that would reveal the outcome in advance. The table therefore passed the required leakage checks before being used in the analysis.

Dim_Material contains 60 materials each assigned a criticality level based on its importance to Keystra's operations:

MATERIAL CRITICALITY	NUMBER OF MATERIALS
Critical	10
High	15
Medium	20
Low	15
<i>Total</i>	<i>60</i>

The material criticality classification gives the analysis an important layer of context. A supplier supporting a critical material carries a different level of potential consequence from one supplying a low-priority material. This distinction becomes particularly important later when supplier risk is connected to material dependencies, assets, and wider operational impact.

Both tables were validated before moving forward. The supplier and material populations were confirmed, predictive or risk leakage was checked, and both datasets were permanently saved and successfully reloaded from storage.

4. FAILURE TYPE DIMENSION

This was created to give the analysis a consistent way of describing how supplier related problems occur. Rather than treating every supplier issue simply as a generic "failure," the model distinguishes between different failure patterns:

- Delivery Failure
- Exposure Related Failure
- Fulfilment Failure
- Multi-Factor Failure
- Quality Failure

The five categories provide a common structure for the later supplier risk labelling and predictive analysis, making it possible to distinguish the nature of a supplier failure and understand what type of problem is contributing to the risk.

5. BRIDGE SUPPLIER MATERIAL

After creating the supplier and material tables, we needed to answer a simple but important question: which supplier provides which material?

That relationship was captured in Bridge_SupplierMaterial. The table contains 174 unique supplier material relationships, covering all 90 suppliers and 60 materials. Every material has at least one qualified supplier, with one supplier identified as the primary supplier for each material.

The sourcing structure was also designed to reflect different levels of dependency:

SOURCING TYPE	MATERIALS
Single Source	7
Dual Source	11
Multi Source	42
<i>Total</i>	<i>60</i>

This matters because the impact of a supplier failure depends partly on whether Keystra has another supplier that can provide the same material. A failure involving a single-source material can therefore create a very different level of exposure from one where alternative suppliers are available.

The relationships, supplier qualifications, dates, and sourcing structures were validated, with no duplicate supplier–material links, invalid references, or predictive leakage detected. The completed bridge was then saved and verified before moving to the next stage of the data model.

6. BRIDGE ASSET MATERIAL

The next relationship we established was between materials and the assets that depend on them. This was necessary to understand what sits behind a supplier risk: if a supplier fails, which assets could actually be affected?

The Bridge_AssetMaterial table contains 141 unique asset material relationships, covering all 13 assets and 60 materials. Each relationship also captures how important that material is to the asset:

REQUIREMENT TYPE	RELATIONSHIPS
Primary	38
Secondary	79
Critical Dependency	24

Total	141
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Across the 13 assets each depends on an average of about 11 materials, while individual materials support between 1 and 3 assets. The table also retains material criticality, allowing the analysis to distinguish ordinary material requirements from dependencies that could have greater operational consequences. This relationship becomes important later in the risk propagation analysis. It allows us to trace the chain from supplier → material → asset, rather than treating supplier failure as an isolated procurement issue.

The bridge passed validation with no duplicate asset material relationships, unresolved material categories, or predictive/risk leakage.

FACTS TABLES DEVELOPMENT

1. FACT PURCHASE ORDER LINE

With the supplier, material, and asset relationships established, the next step was to build the procurement transaction layer. This is where the model starts capturing what actually happened across Keystra's purchasing activity over the analysis period.

Fact_PurchaseOrderLine contains 40,000 purchase order lines across 20,000 purchase orders, covering all 90 suppliers, 60 materials, and 13 assets. The records span the full period from 1 January 2022 to 31 July 2026, giving each supplier a meaningful history for both risk analysis and predictive modelling. The table captures the key elements needed to understand procurement behaviour, including order status, priority, quantities ordered and received, fulfilment, delivery delays, and financial exposure. Some of the key results were:

- * 35,392 lines delivered, with an 87.48% on-time delivery rate.
- * Average fulfilment was 87.24%, with a median of 100%.
- * 4,296 orders were open and 3,650 were partially fulfilled, providing evidence of ongoing fulfilment exposure.
- * Total order value reached approximately \$24.34 billion.
- * Outstanding procurement exposure was approximately \$4.05 billion.

The data was also structured with predictive analysis in mind. All 90 suppliers had observed purchase order history, with an average of approximately 55 months of temporal coverage, providing sufficient historical behaviour from which supplier level features could later be developed.

Before proceeding, the table was checked against the supplier–material and asset–material relationships, as well as quantity, fulfilment, delivery, financial, status, and temporal rules. All foreign keys and required fields were valid, and no predictive or risk leakage was detected.

2. FACT QUALITY EVENT

The procurement data tells us whether orders were delivered and fulfilled, but that alone does not tell us whether the materials received were actually fit for use.

Fact_QualityEvent was therefore added to capture quality problems associated with delivered materials.

The table contains 748 quality events identified from 35,392 delivered PO lines, representing a 2.11% event rate. The events cover all 90 suppliers, 60 materials, and 13 assets, allowing quality issues to be connected back to the wider supply chain.

The recorded events include defects, specification deviations, packaging damage, incorrect materials, documentation non-conformance, and contamination. Severity was also captured, ranging from Low to Critical, together with the quantity affected, quantity rejected, and resolution status. A key result was the relationship between severity and rejection behaviour:

- * Low-severity events → 30.6% mean rejection rate

- * Medium → 43.5%

- * High → 62.1%

- * Critical → 83.1%

Across all events, 23,673 units were affected, with 10,797 rejected, giving an overall rejection rate of 45.61%. Average resolution time was approximately 12 days. Importantly, the quality events were generated using only information that would have been available at the relevant point in time. The current purchase order line was excluded from the supplier's historical performance, and future supplier performance was not used. This preserved the integrity of the later predictive analysis and prevented risk or target leakage. The completed table passed the relationship, temporal, quantity, severity, resolution, and predictive-readiness checks before being locked into the analytical model.

3. FACT INVENTORY SNAPSHOT

The next step was to understand what happens after procurement and delivery: whether the materials Keystra needs are actually available when required. Fact_InventorySnapshot was built as a daily view of inventory for all 60 materials across the 1,673-day analysis

period, resulting in 100,380 material date records. The inventory calculation accounts for procurement demand, actual receipts, rejected quantities, and existing safety stock requirements.

Rather than treating every delivered quantity as usable, quality related rejections were incorporated into the inventory calculation. This means the model reflects usable inventory, not simply what was recorded as received. The resulting inventory position showed:

- * 4,252,719 units of gross receipts
- * 10,797 units rejected
- * 4,241,922 usable units
- * 4,886,270 units of procurement demand
- * 11,596 stockout snapshots, representing 11.55% of all snapshots
- * 13,264 safety-stock breaches, representing 13.21% of snapshots

Each daily material position was also assigned an inventory risk state, ranging from Healthy to Below Safety Stock, Critical, and Stockout. This provides the link between supply-side disruption and the actual availability of materials needed by the business.

The inventory engine was built chronologically, ensuring that future deliveries or future quality events could not influence earlier inventory positions. Validation also confirmed complete material date coverage, consistent stockout and safety-stock logic, non-negative inventory, and valid inventory risk classifications.

ESTABLISHING THE SUPPLIER FAILURE OUTCOME

Once the core supplier, procurement, delivery, material and operational data had been prepared, the focus of the analysis shifted from describing what had already happened to asking a more forward looking question:

could the way a supplier is behaving today provide an early indication that the supplier may become materially unreliable in the future?

To answer that question, Keystra first needed a consistent way of describing a supplier's condition at any point in time. A daily supplier level view was therefore established across the 90 suppliers, and the January 2022 to July 2026 observation period. This allowed each supplier to be assessed using the information that would have been available at that particular point in time, including recent fulfilment and delivery performance, outstanding orders and changes in behaviour relative to its historical performance.

The purpose was not simply to determine whether a supplier had performed well or badly. It was to identify changes in behaviour that could indicate emerging risk.

For example, a supplier that had historically performed reliably but had recently begun experiencing lower fulfilment and increasing delivery delays represents a different risk situation from one whose performance had remained consistently stable. Bringing recent performance together with historical behaviour allowed Keystra to capture this distinction and establish a meaningful baseline of supplier risk.

However, understanding a supplier's current condition was only half of the problem. A predictive model also needs a clearly defined outcome: something that represents the event we are trying to anticipate. Keystra therefore examined what happened to each supplier during the 90 days following every historical observation. This future period was used only to define the outcome, not to describe the supplier at the time the prediction would have been made. In other words, the model would eventually learn from the supplier's condition before the event and use that information to predict what happened afterwards. Maintaining this separation is essential in a time-based predictive problem because allowing future information into the predictors can produce unrealistically strong model performance.

The first attempt at defining supplier failure revealed an important issue. The initial outcome classified 93.85% of observations with a complete future window as failures. This was clearly too broad for a useful predictive system. If almost every supplier observation is considered a failure, the model has little meaningful distinction between a supplier experiencing an ordinary operational problem and one genuinely moving into an unreliable state.

Rather than treating this as a modelling problem, the failure definition itself was reconsidered.

The analysis showed that individual operational problems were too easily being interpreted as evidence of supplier failure. A late delivery, an outstanding order or a fulfilment issue could occur without necessarily meaning that the supplier had entered a materially unreliable condition. Keystra therefore adopted a stricter definition:

A supplier would only be considered to have experienced material failure when its performance deteriorated across at least two independent operational dimensions within the following 90 days. The dimensions covered fulfilment, late delivery, outstanding orders and partial fulfilment.

This changed the question from “Did something go wrong?” to “Did the supplier experience sufficiently broad deterioration to be considered materially unreliable?”. The effect was significant. The future failure rate fell from 93.85% to 7.79%, producing a much more meaningful predictive outcome. Of the 138,953 supplier observations with complete 90-day future outcomes, 10,824 were classified as failures and 128,129 as non-failures.

This was an important point in the development of Keystra because the system was no longer attempting to predict a vague collection of supply chain problems, it now had a specific business outcome representing material supplier unreliability within the next 90 days. The outcome was also checked to ensure that failures reflected the revised definition and that incomplete future periods were not incorrectly treated as non-failures. Most importantly, the information used to define future failure remained separate from the information available at the time of prediction, preserving the temporal structure required for a realistic predictive model. The resulting outcome became the foundation for the predictive analysis.

At this point, Keystra could answer two different but connected questions. It could describe what a supplier looks like today, and it could determine whether that supplier subsequently experienced material deterioration within the next 90 days. The next challenge was to connect these two perspectives. A supplier does not normally move from reliable to failed in a single step. Risk can build gradually as performance weakens, problems persist and multiple warning signals emerge.

Keystra therefore needed to understand not only whether failure occurred, but where each supplier currently sits on the path toward failure. This became the basis for the Supplier Risk-State Framework.

STAGED SUPPLIER RISK FRAMEWORK

Once Keystra had defined what a material supplier failure looks like, the next challenge was to understand how a supplier gets there.

A supplier does not usually go from performing normally to completely failing overnight. There may be a period where performance begins to weaken, followed by a more obvious

deterioration and if the problems continue, a much more serious level of risk. For Keystra to act early, it needed to recognise these stages rather than simply classify suppliers as either “safe” or “failed.”

This led to the development of a staged supplier risk framework that describes a supplier’s current condition based on how its recent performance compares with its own historical behaviour. The system considers changes in fulfilment, partial fulfilment, late deliveries and outstanding orders, allowing it to identify when a supplier is beginning to behave differently from what would normally be expected. The result is a five-stage risk journey:

Stable → Watch → Deteriorating → High Risk → Failure

Each stage represents a different level of concern.

A Stable supplier is operating within its normal range

Watch represents early signs that deserve monitoring

Deteriorating indicates that several aspects of performance are weakening

High Risk represents more severe and sustained deterioration

while Failure is reserved for suppliers experiencing persistent problems across multiple areas of their operations.

This approach makes an important difference to how Keystra understands supplier risk. Instead of simply saying that a supplier is “at risk,” the system can show where that supplier currently sits on the journey and whether its condition is getting worse or improving.

The framework was also designed to reflect the fact that supplier risk can change over time. A supplier showing early warning signs can deteriorate further, but it can also recover if its performance improves. This means the system does not treat risk as a permanent label. It tracks the supplier’s changing condition. The distinction becomes particularly important at the Failure stage. Failure is not triggered by a single poor delivery or isolated operational problem. It represents a sustained deterioration affecting multiple areas of supplier performance. Across the historical data, 9,407 supplier-date observations reached this state, and everyone involved problems at least two independent operational dimensions.

The resulting risk distribution also showed a clear progression in severity. Stable observations had an average risk score of 0.00, compared with 1.01 for Watch, 3.27 for Deteriorating, 9.95 for High Risk and 19.73 for Failure. This increasing separation provides evidence that the states are capturing genuinely different levels of supplier distress rather than simply assigning arbitrary categories.

The supplier histories reinforced this pattern. All 90 suppliers experienced at least a Deteriorating state at some point, 88 progressed to High Risk or above, and 80 eventually reached Failure. More importantly, suppliers did not bypass the deterioration pathway: those that reached Failure had previously passed through High Risk, while those reaching High Risk had previously shown Deteriorating behaviour.

This gives Keystra something the future failure label alone could not provide, which is a view of the supplier's current position and its progression toward risk. The resulting `fact_supplier_risk_state` dataset contains 150,570 supplier-date observations and provides the structured representation of supplier condition needed for the next stage of the system.

At this point, Keystra could answer two connected questions:

What condition is the supplier in now?

and

Is that condition associated with a material failure in the future?

The next step was to bring these signals together into a model-ready dataset, allowing Keystra to move from describing supplier risk to predicting which suppliers are most likely to deteriorate in the future.

TURNING SUPPLIER RISK SIGNALS INTO A PREDICTIVE DATASET

With the supplier risk journey established, Keystra now had a structured view of how supplier conditions develop over time. The next challenge was to turn that information into something a predictive model could actually learn from.

The key principle was simple:

The model must make its prediction using only information that would have been available at the time. If Keystra wants to know whether a supplier is likely to fail in the next 90 days, it cannot use information about what happened during those 90 days to make the prediction.

For each supplier and observation date Keystra therefore brought together the information available at that point in time. This included recent supplier performance, longer term performance history, procurement activity, outstanding orders, fulfilment and delivery behaviour, current risk signals, risk state and changes in performance relative to the supplier's normal pattern.

The inclusion of these changes was particularly important. Looking only at a supplier's current performance can miss the difference between a supplier that has always operated at a certain level and one whose performance has recently started to deteriorate. Keystra

therefore captured both the supplier's current position and the direction in which that position was moving. The future failure outcome developed earlier was then used as the target the model would learn to predict. In practical terms, each historical observation asks the same question Keystra would eventually ask in a live environment:

Based on what we knew about this supplier at this point in time, did the supplier go on to experience a material failure within the following 90 days?

This required a strict separation between the information used to make the prediction and the information used to determine the eventual outcome. Future failure information was therefore kept out of the predictive features, ensuring that the model was learning from the conditions that existed before failure rather than simply recognising evidence that appeared afterwards. Observations that did not have a complete 90-day period available for determining the outcome were also excluded from modelling. This left 138,953 usable supplier-date observations across all 90 suppliers, with 7.79% subsequently experiencing a material failure.

The resulting dataset brought together 54 predictive features representing supplier activity, recent and historical performance, exposure, operational risk signals, deterioration patterns and current risk condition. Rather than relying on a single indicator, the model could therefore learn from multiple aspects of supplier behaviour and identify combinations of signals that may precede future failure.

At this point, the problem had fundamentally changed. Keystra was no longer simply describing supplier performance or assigning suppliers to risk states. It had created a historical record of what suppliers looked like before their outcomes were known, together with a clearly defined measure of what happened afterwards. The remaining challenge was to determine whether a model could learn this relationship reliably. Because supplier conditions changed substantially over the analytical period, simply mixing all historical observations together would not provide a realistic test. Keystra therefore moved to a chronological modelling approach, using earlier supplier behaviour to train the model, a later period to validate it, and the most recent period to test how well it could identify future failures.

This established the foundation for the predictive modelling stage, learning from the past to identify suppliers that may become materially unreliable in the future.

TURNING SUPPLIER RISK SIGNALS INTO A PREDICTIVE EARLY WARNING MODEL

At this stage of Keystra we had done the groundwork needed to answer a more important question. We knew how to describe a supplier's current condition, the signals that could indicate deterioration, and also defined what it meant for a supplier to

experience a material failure within the next 90 days. But there was still one thing we had not proven:

Could Keystra use what it knows about a supplier today to predict what is likely to happen to that supplier tomorrow?

That became the focus of the predictive modelling stage. The first step was to create a baseline model. The purpose of a baseline is simple: before trying to build something better, we need to know how well a reasonable starting point can perform.

Keystra used a Random Forest as this starting model. It was trained using the supplier activity, recent performance, historical behaviour, exposure, risk signals, deterioration indicators and current risk state developed in the previous stages. The data was divided according to time. Earlier observations were used to train the model, a later period was used to validate it, and the most recent period was kept completely unseen for the final test. This was important because Keystra is supposed to predict the future.

We therefore wanted to test the model in the same way it would eventually be used in practice: learn from the past, then predict what happens later.

The baseline model performed well enough to prove that the supplier data contained useful predictive information. On the unseen test period, the Random Forest achieved 86.13% precision, 66.45% recall, and a PR-AUC of 0.8878. The precision result was encouraging.

When the model warned that a supplier was likely to experience future failure, it was correct most of the time. However, recall exposed a problem;

Recall tells us how many of the failures that actually happened were successfully detected by the model. The baseline recall of 66.45% meant that the model detected 4,613 of the 6,942 future failures, but missed 2,329.

That is roughly one out of every three future failures going undetected. For a normal classification problem, that might be acceptable depending on the application.

For Keystra, it was not ideal. The whole purpose of an early-warning system is to give the business an opportunity to act before a supplier becomes materially unreliable. If too many genuine failures are missed, the system is not providing enough protection. So the question became:

Can we make the model better at catching future failures without simply creating a huge number of false alarms?

First Attempt: Adjusting the Warning Threshold. Before changing the model itself, Keystra tested whether the problem could be improved by changing the point at which a prediction becomes a warning. The model produces a probability for each supplier observation. For example, it might estimate that a supplier has a 70% probability of future

failure. A threshold determines whether that probability is high enough for Keystra to issue a warning. A lower threshold means the system becomes more cautious and flags more suppliers. This can help catch more failures, but it can also create more unnecessary warnings. A higher threshold does the opposite. The system becomes more selective, but it may miss more genuine failures.

Keystra therefore tested different thresholds using the validation data. A threshold of 0.80 appeared promising. It produced 87.19% recall and 74.84% precision on the validation period. At first, this looked like a significant improvement, but there was an important test still to come. The 0.80 threshold was frozen and applied to the completely unseen test period. Recall then fell to only 36.20%.

That result was extremely useful because it showed that changing the threshold was not enough. Why? Because the supplier environment had changed considerably over time. The Supplier Environment Was Changing

Failure rates were not constant throughout the dataset.

The training period had a failure rate of 2.07%.

The validation period increased to 8.98%.

The final test period reached 32.89%.

In other words, the later supplier environment contained far more failures than the earlier environment on which the model had learned. This is known as temporal drift. In simple terms, the world the model was learning from was changing. A model trained mainly on a period where failures were relatively uncommon may struggle when the underlying supplier environment becomes much more failure-prone. This was important for Keystra because a real supply-chain risk system cannot assume that supplier behaviour will always remain the same. Conditions can change, and the model needs to be able to recognise increasingly complex patterns of risk.

The threshold experiment therefore gave us an important conclusion: The problem was not simply where we placed the warning line, we also needed to improve the model itself.

Keystra therefore compared different modelling approaches rather than relying entirely on the original Random Forest. The original Random Forest remained the baseline, an improved Random Forest was then tested to determine whether the same model family could perform better after refinement.

Finally, a Histogram Gradient Boosting model was tested as an alternative approach. The purpose of comparing these models was not to use multiple models simply for the sake of complexity, each comparison answered a practical question:

Can another modelling approach learn the relationship between supplier behaviour and future failure better than our baseline? The answer was yes.

On the validation period:

- * The original Random Forest achieved a PR-AUC of 0.8795
- * The improved Random Forest achieved 0.9057
- * Histogram Gradient Boosting achieved 0.9482

Histogram Gradient Boosting was therefore selected as the strongest candidate for the final predictive model. The reason for using PR-AUC as an important comparison measure was straightforward, Keystra is particularly interested in finding future failures while avoiding an overwhelming number of false warnings. Precision-Recall performance therefore tells us more about the usefulness of the model for this specific problem than accuracy alone.

Even after selecting the stronger model, Keystra still needed to decide when a predicted probability should become an actual warning. The threshold was selected using the validation data, not the test data. This is important because the test data must remain untouched until the very end. Otherwise, we would effectively be using the answers from the test to help build the system we later claim to be testing. A threshold of 0.30 was selected for the Histogram Gradient Boosting model.

At this threshold, the model achieved:

- * 93.65% recall
- * 84.21% precision
- * 88.68% F1-score

on the validation period. The threshold was then frozen, only after that was the model evaluated on the completely unseen test period.

The final Histogram Gradient Boosting model produced a substantial improvement over the original Random Forest baseline.

Final Model Performance Table

METRIC	RANDOM FOREST BASELINE	FINAL HISTOGRAM GRADIENT BOOSTING
Precision	86.13%	95.27%
Recall	66.45%	87.05%
F1 Score	75.02%	90.97%
PR-AUC	0.8878	0.9721

ROC-AUC	0.9598	0.9893
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The most important result was the improvement in recall. The final model correctly detected 6,043 of the 6,942 observations that subsequently experienced material failure, it missed 899. At the same time, it generated only 300 false warnings among observations that did not subsequently result in failure. This means that the final model detected 87.05% of future failures while maintaining 95.27% precision. That is a major improvement over the baseline. The baseline detected about two-thirds of future failures. The final model detected almost nine out of every ten. Just as importantly, precision also improved. The final model was not simply catching more failures by warning about almost everybody, it was becoming better at distinguishing genuine future failures from suppliers that would remain broadly reliable.

The model's feature importance results also helped explain what was driving its predictions. The strongest signals were related mainly to recent supplier activity and fulfilment behaviour. The most influential features included:

- * Rolling 90-day purchase order activity
- * Fulfilment risk
- * Recent 90-day fulfilment rate
- * Partial fulfilment activity
- * Outstanding quantities
- * Recent delivery delays
- * Deviation from historical fulfilment behaviour

This is important because it gives Keystra a business interpretation of the model. The model is not simply looking at whether a supplier has ever performed badly, it is looking for signs that something is changing. For example, a supplier that has historically been reliable but suddenly begins receiving more purchase orders, fulfilling fewer quantities, accumulating outstanding orders and delivering later than usual may represent a different risk from a supplier whose performance has remained stable. This is exactly the type of behaviour this solution was designed to detect.

The predictive modelling stage therefore established something fundamental; Supplier behaviour contains measurable signals that can provide advance warning of future material failure.

The Random Forest baseline proved that the predictive problem was viable. The threshold analysis showed that simply changing the warning level could not fully solve the problem, particularly in an environment where failure behaviour was changing over time.

Comparing alternative models then showed that Histogram Gradient Boosting could capture the predictive relationships more effectively.

The final model achieved 87.05% recall, 95.27% precision, 0.9721 PR-AUC and 0.9893 ROC-AUC on completely unseen test data. At this point, Keystra could answer:

“How likely is this supplier to experience material failure within the next 90 days?”

But a probability alone is not enough for an operational risk system, a procurement manager does not just need to know that Supplier A has a 94% failure probability. They need to know how serious the warning is, which suppliers should be addressed first, what is driving the risk, and where intervention may be most valuable. That is the next step. The predictive model now becomes the engine behind the Supplier Early Warning and Risk Prioritisation System, where predictions are transformed into supplier level warnings, severity levels, risk rankings and explainable risk drivers.

TURNING SUPPLIER FAILURE PREDICTIONS INTO ACTIONABLE RISK WARNINGS

Knowing that a supplier is likely to fail is useful, but it is not enough. A procurement or operations team still needs to know which suppliers require attention first, why they are being flagged, and what kind of response may be appropriate. This was the purpose of the early-warning layer.

The predictive model provided each supplier with a probability of experiencing material failure within the next 90 days. The early-warning engine then turned those probabilities into a current supplier risk view by combining the prediction with the supplier’s existing risk condition, deterioration signals and evidence of problems across multiple operational dimensions. Using the latest available modelling date of 30 July 2026, Keystra evaluated 88 suppliers with complete current information. A supplier was considered to have an active early warning when its predicted failure probability exceeded the operating threshold of 0.30, which had been selected during model validation. This resulted in 63 of the 88 suppliers being flagged for attention.

However, treating all 63 suppliers as equally urgent would not make sense. A supplier with an elevated probability but limited evidence of deterioration requires a different response, from one that is already experiencing persistent problems across several areas. The system therefore assigned each warning a severity level and priority based on the combination of predicted failure probability and current supplier condition. The resulting population contained:

- * 31 Critical warnings
- * 32 High warnings
- * 5 Moderate warnings

*** 20 Low warnings**

This created a more useful risk picture than simply producing a list of suppliers with high predicted probabilities. It allowed Keystra to distinguish between suppliers that may require immediate intervention and those that should primarily be monitored.

The next question was why a supplier was being flagged? The system therefore evaluated 24 potential risk drivers and identified the strongest drivers for each supplier. These drivers included delivery delays, late deliveries, partial fulfilment, outstanding quantities, recent order activity, deterioration from historical performance, persistent deterioration and problems occurring across multiple operational dimensions.

This was important because suppliers do not necessarily become risky in the same way. For example, S035 the highest priority supplier had a 97.68% predicted probability of failure. The supplier was already classified as Failure, with four severe signals and problems across three independent operational dimensions. Its strongest risk drivers included risk state severity, multiple failure dimensions, average delivery delay and outstanding quantity exposure. The appropriate response was therefore a supplier performance review combined with outstanding-order recovery.

Another supplier, S025, had an even higher predicted failure probability of 99.31%, but its dominant risk pattern was different. Its strongest drivers were deterioration in partial fulfilment, recent partial fulfilment and increasing partial-fulfilment activity. In this case, the warning points more directly toward a fulfilment-performance problem rather than primarily a delivery-delay problem.

This distinction is what makes the early-warning layer useful. The system is not simply saying:

"This supplier is risky."

It is moving toward:

"This supplier is risky because these specific aspects of its behaviour are deteriorating, and this is the type of intervention that should be considered."

The same logic can be seen across the highest priority suppliers. Some were driven primarily by delivery delays, others by partial fulfilment, outstanding quantities or persistent deterioration. This means the system can recognise different pathways through which supplier risk is developing rather than treating supplier failure as a single uniform event.

The predictive probability, risk state and risk drivers therefore serve different purposes.

The failure probability answers:

How likely is this supplier to experience material failure within the next 90 days?

The risk state answers:

What condition is the supplier currently in?

The risk drivers answer:

What is contributing to that condition and prediction?

The priority ranking answers:

Which suppliers should receive attention first?

And the recommended intervention provides an initial indication of how the identified risk could be addressed. Together, these elements turn the machine-learning output into something much closer to an operational risk tool.

The final fact_supplier_explainable_warning object contains 88 current supplier records and 109 analytical attributes, with explanations generated for all 88 suppliers and warnings identified for 63. It brings the predictive probability, warning severity, risk priority, current and previous risk states, risk-state transition, key risk drivers and recommended intervention into a single supplier-level view. This represents an important shift in the Keystra system.

The earlier stages established what supplier failure means, what supplier deterioration looks like, and which suppliers are likely to experience future failure. This layer makes those predictions understandable and actionable, Keystra can now move beyond:

“Which suppliers are likely to fail?”

to:

“Which suppliers need attention, why are they at risk, and what should we investigate?”

The remaining gap is context. A supplier warning becomes far more important when Keystra knows what that supplier provides, which materials are affected, which assets depend on those materials, how critical those materials are, and what operational consequences could follow if supply is disrupted. That is the next stage of the system: connecting supplier risk to the wider supply chain and the operations that depend on it.

By this stage of the Keystra Supply Chain Risk & Operational Resilience system, the analysis had moved well beyond simply identifying suppliers that were performing poorly.

The predictive layer had established which suppliers were likely to become unreliable. The explainability layer then addressed the next question: why were these suppliers being flagged? But there was still a more important business question:

If a supplier becomes unreliable, how much does it actually matter to Keystra?

A supplier can have a high probability of failure without necessarily creating a major operational problem. Another supplier may have a lower probability of failure but sit at the centre of several critical dependencies. What matters therefore is not only whether a supplier may fail, but what sits behind that supplier, and what could be affected if the risk materialises. This is where the analysis moved from supplier-level risk into supply-chain context.

Understanding Why the Risk Matters

The explainability layer connected each supplier's predictive risk to the underlying factors driving that risk. Across the 90-supplier population, 88 suppliers had sufficient predictive or risk-based intelligence to generate an active warning, while two suppliers had no current predictive warning available. For the suppliers with active intelligence, the system did not simply attach a risk label. It decomposed the risk into its major drivers and translated those drivers into a business interpretation.

Failure probability was the dominant primary driver, accounting for 62 suppliers, or 68.89% of the population. Supply chain exposure was the primary driver for another 27 suppliers, while intervention urgency was the primary driver for one supplier. This distinction is important. For some suppliers, the main concern was that the supplier itself appeared to be moving towards an unreliable state. For others, the greater concern was the level of dependency Keystra had already built around that supplier.

The root-cause analysis therefore allowed Keystra to distinguish between different situations rather than treating every supplier alert as the same type of problem.

SunCore Energy Technologies Ltd. illustrates this clearly. The supplier had an estimated failure probability of approximately 99%, a supplier risk priority score of 83.04, a supply chain exposure score of 100 and an intervention urgency score of 99.64. The concern was therefore not simply that SunCore was predicted to fail, the concern was that a highly probable supplier failure was connected to highly important parts of the Keystra supply chain.

Process Materials Solutions Ltd., and Sterling Industrial Equipment Ltd. showed a similar pattern. Process Materials Solutions had a predicted failure probability of approximately 99% alongside a supply chain exposure score of 93.75, while Sterling Industrial Equipment had approximately 92% predicted failure probability and a supply-chain exposure score of

100. These examples demonstrate why explainability is essential. A risk score tells management how much risk exists, while the explanation begins to show why the risk exists and matters. However, understanding the driver of the risk was only part of the problem, Keystra also needed to understand what that supplier was connected to.

Tracing Supplier Risk Through the Supply Chain

The risk propagation layer connected suppliers to the operational network through the materials and assets dependent on them. The resulting dependency chain was:

Supplier → Material → Asset → Business Unit

This created a traceable view of how supplier failure could move beyond procurement and into actual operations. The network covered 90 suppliers, 60 materials, 13 assets and all five Keystra business units, with 407 unique supplier-material-asset relationships. The underlying purchase-order population contained 40,000 lines, representing 622,917 outstanding units and approximately \$4.05 billion in outstanding procurement exposure.

A critical control was applied at this stage: the propagated exposure had to reconcile to the underlying procurement exposure. It did.

The underlying exposure was approximately \$4.050 billion, and the final propagated exposure was also approximately \$4.050 billion, with an exposure reconciliation ratio of 1.00 and no orphan dependency links. This matters because the propagation engine was not creating a new or inflated financial exposure. It was taking the exposure that already existed and showing where that exposure sat within the operational dependency network. The resulting analysis identified 20 suppliers with Critical Propagation risk, 27 with High Propagation risk and 33 with Moderate Propagation risk.

SunCore Energy Technologies Ltd. again provides a useful example. Its propagation score was 94.92, with a 99% predicted failure probability. The supplier was connected to three materials, including one critical material, five assets, all five classified as critical, four business units and approximately \$31.6 million in outstanding exposure. That changes the interpretation of the supplier completely. SunCore is not simply a supplier with a high failure probability, it is a supplier whose potential failure sits within a network of important operational dependencies.

The same principle appeared elsewhere in the network. DeltaFlow Technologies Ltd. for example carried approximately \$456.2 million in exposed outstanding value, while other suppliers with much smaller financial exposure achieved similarly high propagation scores because they were connected to more critical operational dependencies. This showed why a supplier-risk model based purely on procurement value would be insufficient. Financial exposure tells Keystra how much is committed. Dependency tells Keystra what could be affected. The propagation layer therefore transformed supplier risk into operational risk.

From Risk Assessment to Intervention

Once the supplier risk had been explained and its potential consequences had been mapped, the next question became much more practical:

What should Keystra actually do about it?

The intervention engine brought the different risk signals together to determine how seriously each supplier should be treated. The objective was not to rank suppliers simply by their probability of failure. Instead, the system considered the combination of:

- * Predicted supplier failure probability
- * Supplier risk priority
- * Supply-chain exposure
- * Critical material dependency
- * Asset dependency
- * Business impact
- * Intervention urgency

This allowed Keystra to distinguish between a supplier that is becoming unreliable and a supplier whose deterioration could have significant consequences for the organisation. Across the 90 suppliers, the intervention assessment classified:

Intervention Level	Suppliers
Immediate intervention	16
High-priority intervention	36
Moderate intervention	27
Monitor	11

This means that 52 of the 90 suppliers require active intervention, while 11 can remain under monitoring based on their current risk state. The distinction is important because intervention should be proportional to risk.

Where high failure probability is combined with significant operational exposure, the appropriate response becomes immediate supplier stabilisation, recovery planning, protection of critical materials, contingency sourcing and management escalation.

Where the main concern is dependency rather than imminent failure, Keystra can focus on alternative sourcing, reducing supplier concentration and strengthening contingency arrangements.

Where risk is less severe, enhanced monitoring and structured mitigation may be sufficient.

This is particularly important for suppliers such as NexControl Engineering Ltd. and CoreProcess Materials Ltd. Their predicted failure probabilities were below 50%, yet both had supply-chain exposure scores of 100. A probability only model could easily push these suppliers down the priority list, but the intervention logic does not. It recognises that lower likelihood does not automatically mean lower business risk when the consequence of failure is significant. This is the point where the system moves from prediction to management judgement. It is no longer asking only:

“Who is likely to fail?”

It is asking:

“Who could cause the greatest problem if they fail, and who should Keystra deal with first?”

Turning Risk into an Early Warning

An intervention assessment describes what Keystra should do based on the current risk state. But supplier risk is not static, a supplier classified as Moderate today could become High tomorrow. A supplier with manageable exposure could become critical as outstanding orders accumulate or inventory protection decreases. A supplier showing early performance deterioration could also move towards a much higher predicted probability of failure. Keystra therefore needed a mechanism that could translate these changing conditions into a clear warning that risk owners could immediately understand and act upon. This led to the final early warning layer.

The early warning engine takes the supplier’s predictive risk, supply chain exposure, intervention urgency and downstream impact and converts them into a defined warning severity, risk type, explanation, recommended response and management escalation. Across the complete population of 90 suppliers, the final warning distribution was:

ALERT SEVERITY	SUPPLIERS	PERCENTAGE
Critical	10	11.11%
High	26	28.89%
Moderate	35	38.89%
Low	17	18.89%
No Current Warning	2	2.22%

The ten Critical suppliers represent the most urgent cases. They require action within 0–7 days and escalation to Executive or Supply Chain Leadership.

The 26 High-severity suppliers require action within 14 days and are escalated to Supply Chain Risk Management.

This gives management something much more useful than a table of model scores. Instead of asking someone to review 90 suppliers and decide which ones deserve attention, the system has already separated the population according to the urgency of the situation. The Critical population again highlights the value of combining prediction with consequence.

SunCore Energy Technologies Ltd. recorded the highest early-warning severity score at 95.34. Process Materials Solutions Ltd. followed at 92.96, while Sterling Industrial Equipment Ltd. recorded 91.86. These suppliers are not being flagged simply because the model produced a high probability of failure. Their warnings are strengthened by the exposure and operational dependency surrounding them.

The warning engine also distinguishes between different types of risk. Across the supplier population, 27 suppliers generated Predictive Supplier Failure Alerts, 18 generated Emerging Supplier Risk Alerts, 8 generated Supply Chain Exposure Alerts and 4 generated combined Predictive Supplier Failure and Exposure Alerts. Another 31 suppliers remained within general Supplier Risk Monitoring. This distinction allows the response to reflect the nature of the problem. A supplier approaching failure requires a different response from a supplier whose primary concern is unusually high operational exposure.

The two suppliers without a current predictive warning, PrimeFlow Industrial Solutions Ltd. and WindSun Energy Systems Ltd., were deliberately not assigned an artificial low-risk classification. Instead, they were retained as No Current Predictive Warning cases, with refreshed predictive intelligence recommended before a material risk decision is made. This is an important control because no warning does not mean no risk, it means there is currently insufficient predictive evidence to issue a stronger warning.

What Keystra Can Now Do

At the end of this analytical chain, the supplier risk system has become considerably more than a predictive model. It can now move through the full risk journey:

Supplier Risk



Predicted Supplier Failure



Risk Explanation & Root Cause



Material Dependency



Asset & Business Exposure



Supply-Chain Impact



Intervention Priority



Early-Warning Severity



Recommended Response



Management Escalation



Action

This is the real value of the final analytical layer. Keystra does not simply identify a supplier that may fail. It can explain why the supplier is being flagged, identify what that supplier supports, determine how significant the downstream exposure could be, establish how urgently the organisation should respond and translate the assessment into a practical management action. The distinction can be expressed simply:

- Prediction tells Keystra who may fail.
- Explainability tells Keystra why.
- Propagation tells Keystra what could be affected.
- Prioritisation tells Keystra who matters most.
- Early warning tells Keystra when attention is required and what response should follow.

That progression is what turns supplier analytics into supply-chain risk management. The final system therefore changes the question from: “Which suppliers are risky?” to: “Which suppliers are becoming unreliable, why are they at risk, what could their failure affect, how urgently should Keystra respond, and what action should be taken?”

The Final Supplier Risk Register: 'From 90 Suppliers to Clear Management Priorities'

Everything that has been built throughout the Keystra supply-chain risk journey ultimately comes together in the final supplier risk register.

The purpose of this register is not simply to produce another ranking of suppliers. It is the final expression of everything the system has learned about the supplier population. Each supplier has been assessed not only on the likelihood that it may enter an unreliable state, but also on what that supplier is connected to, the potential exposure created by that dependency, the extent to which risk could propagate through the supply chain, and the level of attention that should be given to the supplier.

The result is a complete, explainable classification of all 90 suppliers. Each supplier now has a defined position within the Keystra risk hierarchy, together with its Early Warning Score, intervention priority and primary risk driver. This means that the final list does not simply tell management which suppliers are ranked highest. It provides context for why they are ranked where they are. The final supplier population is distributed as follows:

RISK CLASSIFICATION	SUPPLIERS	SHARE OF SUPPLIERS
Critical Early Warning	1	1.11%
High Early Warning	10	11.11%
Emerging Risk	21	23.33%
Watchlist	34	37.78%
Low Current Risk	24	26.67%

This distribution immediately shows that risk is not evenly distributed across the supplier base. Only 11 suppliers representing 12.22% of the total population fall into the Critical or High Early Warning categories. Yet these 11 suppliers account for approximately \$1.70 billion in outstanding exposure, representing 42.01% of the total \$4.05 billion exposure across the supplier population. That concentration is one of the strongest findings from the entire analysis, showing that Keystra does not have 90 suppliers requiring the same level of attention. Instead, the system has identified a relatively small group carrying a disproportionately large share of the potential exposure.

This allows management and risk owners to focus their resources where they can have the greatest impact, while still maintaining appropriate monitoring across the wider supplier population.

The Highest-Risk Supplier

At the very top of the final register is S004 — Sterling Industrial Equipment Ltd.

Sterling Industrial Equipment Ltd. received an Early Warning Score of 77.5/100, making it the only supplier classified as a Critical Early Warning and the number-one supplier in the final risk ranking. Its predicted supplier-failure probability is 92%, while its downstream propagation score is 73.14/100. Most significantly, the supplier has approximately \$1.258 billion in outstanding exposure.

This combination is what makes S004 particularly important. The concern is not simply that Sterling Industrial Equipment Ltd. has a high probability of entering an unreliable state, the greater concern is the scale of Keystra's existing exposure to the supplier, and the potential consequences if that exposure becomes disrupted. Its primary risk driver is therefore Financial Exposure.

The system assigns S004 an Immediate intervention priority. The appropriate management response is immediate supplier intervention, protection of critical dependencies, contingency planning and leadership escalation. This is an important example of how the final ranking should be interpreted. S004 is number one not because it has the highest predicted failure probability in the population. Several other suppliers have probabilities of 99%, but it is number one because the system considers likelihood together with consequence. That distinction is fundamental to risk analytics.

A supplier that is highly likely to fail but has limited exposure, may require a different response from a supplier whose deterioration could place more than a billion dollars of outstanding procurement exposure at risk. The final register captures that difference.

The High Early Warning Group

Immediately below S004 are ten suppliers classified as High Early Warning.

These suppliers are:

RANK	SUPPLIER	SCORE	PRIMARY RISK DRIVER
2	S054_ Industrial Supply Partners Ltd	72.9	Business Unit Reach
3	S055_ Process Materials Solution Ltd	71.6	Business Unit Reach
4	S087_ Continental Industrial Services Ltd	70.9	Supplier Failure Probability
5	S058_ Nexus Industrial Supply Ltd	70.5	Business Unit Reach
6	S081_ Suncore Energy Technologies Ltd	68.9	Supplier Failure Probability

7	SO19_Powerlink Industrial Systems Ltd	67.4	Business Unit Reach
8	SO25_Brightline Electrical Industries Ltd	67.4	Business Unit Reach
9	SO35_Advanced Process Instruments Ltd	66.6	Supplier Failure Probability
10	SO33_ProSense Industrial Controls Ltd	66.5	Critical Material Exposure
11	SO18_Crown Electrical Technologies Ltd	65.9	Business Unit Reach

What is important here is the variation in the drivers.

Industrial Supply Partners, Process Materials Solutions, Nexus Industrial Supply, Powerlink Industrial Systems, Brightline Electrical Industries and Crown Electrical Technologies are primarily influenced by Business Unit Reach. Their risk is therefore strongly connected to the breadth of the organisation's dependency on them.

Continental Industrial Services, SunCore Energy Technologies and Advanced Process Instruments are primarily driven by Supplier Failure Probability. In these cases, the supplier's own predicted deterioration is a major component of the risk assessment.

ProSense Industrial Controls is different again. Its primary driver is Critical Material Exposure, meaning the importance of the materials connected to the supplier materially increases the consequence of its potential deterioration.

This is precisely why the final register is more useful than a simple probability ranking. Ten suppliers may all require high-priority attention, but they do not necessarily require the same management response.

One of the clearest demonstrations of the value of the final scoring framework can be seen when the predicted failure probabilities are compared with the final rankings. Several suppliers have predicted failure probabilities close to or equal to 99%.

For example:

S055 — Process Materials Solutions Ltd. has a predicted failure probability of 99%.

S087 — Continental Industrial Services Ltd. has a predicted failure probability of 99%.

S058 — Nexus Industrial Supply Ltd. has a predicted failure probability of 99%.

S081 — SunCore Energy Technologies Ltd. has a predicted failure probability of 99%.

S035 — Advanced Process Instruments Ltd. has a predicted failure probability of 98%.

S018 — Crown Electrical Technologies Ltd. has a predicted failure probability of 97%.

Yet these suppliers do not all occupy the same position simply because their failure probabilities are similar. SunCore Energy Technologies Ltd., for example, has a failure probability of 99% and an exceptionally high propagation score of 94.92/100. Its risk is therefore reinforced by the extent to which its deterioration could move through the wider supply-chain network. ProSense Industrial Controls Ltd. presents a different situation. Its predicted failure probability is only 64%, substantially lower than many suppliers above it, but its propagation score is 81.63/100 and its outstanding exposure is approximately \$58.77 million. Its primary risk driver is Critical Material Exposure.

The final ranking therefore captures something that a failure-probability model alone cannot. It asks not only:

“How likely is this supplier to become unreliable?”

It also asks:

“What happens to Keystra if it does?”

That is the difference between predicting supplier failure and understanding supplier risk.

Emerging Risk: The Suppliers Keystra Should Not Ignore

Below the Critical and High groups are 21 Emerging Risk suppliers. These suppliers are not currently in the highest intervention category, but they are important because they represent the population in which risk may develop into something more serious if the underlying conditions worsen. The Emerging Risk group begins with:

S017 — VoltEdge Engineering Ltd. | Score: 64.3 | Critical Asset Exposure

S013 — Westfield Industrial Engineering Ltd. | Score: 63.5 | Supplier Failure Probability

S089 — Nexus Energy Equipment Solutions Ltd. | Score: 63.4 | Supplier Failure Probability

S009 — Ironbridge Equipment Solutions Ltd. | Score: 62.7 | Supplier Failure Probability

S047 — DeltaFlow Technologies Ltd. | Score: 61.7 | Downstream Propagation

S001 — Apex Rotating Systems Ltd. | Score: 59.6 | Supplier Failure Probability

S030 — Axis Process Controls Ltd. | Score: 58.7 | Supplier Failure Probability

S003 — Atlas Turbomachinery Services Ltd. | Score: 58.2 | Supplier Failure Probability

S010 — Pinnacle Machinery Services Ltd. | Score: 57.3 | Supplier Failure Probability

S024 — Enertech Power Solutions Ltd. | Score: 56.4 | Supplier Failure Probability

S086 — Integrated Energy Supply Solutions Ltd. | Score: 56.0 | Business Unit Reach
S069 — West Africa Chemical Solutions Ltd. | Score: 54.7 | Supplier Failure Probability
S044 — Precision Valve Systems Ltd. | Score: 53.4 | Business Unit Reach
S061 — MetroProcess Supplies Ltd. | Score: 53.3 | Supplier Failure Probability
S007 — Prime Industrial Machinery Ltd. | Score: 53.2 | Downstream Propagation
S090 — Frontier Industrial Technologies Ltd. | Score: 53.1 | Supplier Failure Probability
S079 — SolarEdge Industrial Energy Ltd. | Score: 51.1 | Supplier Failure Probability
S063 — General Process Solutions Ltd. | Score: 50.5 | Downstream Propagation
S076 — Sentinel Safety Equipment Ltd. | Score: 50.5 | Supplier Failure Probability
S015 — Grandview Mechanical Systems Ltd. | Score: 50.4 | Supplier Failure Probability
S040 — FlowGuard Industrial Ltd. | Score: 50.3 | Supplier Failure Probability

The purpose of this group is not to suggest that these suppliers are already critical, but to make sure Keystra sees the risk before it reaches that point. For example, DeltaFlow Technologies Ltd. has a relatively high downstream propagation score and approximately \$456.2 million in outstanding exposure. Its primary risk driver is Downstream Propagation. That means its importance comes not simply from the probability of supplier deterioration, but from what could happen across the connected supply chain if that deterioration occurs. VoltEdge Engineering Ltd. presents another type of situation, with Critical Asset Exposure as its primary driver. These differences matter because they tell risk owners where to investigate. An emerging supplier failure risk may require closer monitoring of supplier performance and financial or operational indicators. A downstream propagation risk may require contingency planning around dependent materials and assets. A critical asset exposure may require stronger protection of the operational assets connected to the supplier.

The Emerging Risk category therefore acts as an early intervention layer. It is where Keystra can still act before a developing risk becomes a high-priority disruption.

The Watchlist Population

The largest group in the final classification is the Watchlist, containing 34 suppliers. These suppliers are not currently considered high enough risk to require immediate or planned intervention, but they should not disappear from the risk management process.

The Watchlist includes suppliers such as:

S011 — BluePeak Mechanical Systems Ltd. | 49.9 | Downstream Propagation

S021 — PrimeVolt Engineering Services Ltd. | 49.9 | Downstream Propagation

S062 — Reliable Industrial Materials Ltd. | 49.4 | Supplier Failure Probability

S082 — RenewPower Engineering Ltd. | 49.2 | Supplier Failure Probability

S006 — Crestline Rotating Equipment Ltd. | 49.0 | Downstream Propagation

S088 — Prime Integrated Engineering Ltd. | 49.0 | Business Unit Reach

S064 — PetroChem Solutions Ltd. | 49.0 | Supplier Failure Probability

S083 — HelioGrid Solutions Ltd. | 48.6 | Downstream Propagation

S059 — Westbridge Process Supplies Ltd. | 48.3 | Business Unit Reach

S071 — ProcessChem Technologies Ltd. | 48.2 | Supplier Failure Probability

S080 — GreenPeak Renewable Systems Ltd. | 47.6 | Downstream Propagation

S038 — OptiSense Systems Ltd. | 47.3 | Supplier Failure Probability

S032 — ClearPoint Instrumentation Ltd. | 47.1 | Downstream Propagation

S008 — Vector Mechanical Engineering Ltd. | 45.6 | Supplier Failure Probability

S014 — Orion Machinery & Engineering Ltd. | 44.2 | Supplier Failure Probability

S031 — Integra Automation Systems Ltd. | 44.0 | Business Unit Reach

S027 — Westbridge Power Systems Ltd. | 43.8 | Business Unit Reach

S036 — ControlSphere Technologies Ltd. | 43.6 | Business Unit Reach

S022 — Northbridge Power Equipment Ltd. | 43.1 | Supplier Failure Probability

S034 — NexControl Engineering Ltd. | 43.1 | Downstream Propagation

S026 — VectorGrid Technologies Ltd. | 42.9 | Supplier Failure Probability

S029 — SignalWorks Instrumentation Ltd. | 42.7 | Downstream Propagation

S057 — CoreProcess Materials Ltd. | 42.6 | Downstream Propagation

S043 — HydroFlow Engineering Ltd. | 41.4 | Business Unit Reach

S060 — PrimeSource Industrial Ltd. | 39.1 | Downstream Propagation

S065 — LubriTech Industrial Ltd. | 38.9 | Supplier Failure Probability

S046 — IronGate Piping Systems Ltd. | 38.0 | Downstream Propagation

S084 — EcoVolt Renewable Equipment Ltd. | 37.9 | Business Unit Reach

S039 — CoreLogic Instrumentation Ltd. | 37.2 | Downstream Propagation

S053 — Continental Piping Technologies Ltd. | 37.0 | Business Unit Reach

S023 — Arcstone Electrical Systems Ltd. | 36.6 | Business Unit Reach

S042 — Pipeline Dynamics Ltd. | 36.5 | Downstream Propagation

S037 — Meridian Automation Solutions Ltd. | 35.6 | Business Unit Reach

S016 — GridCore Power Systems Ltd. | 35.1 | Downstream Propagation

This group demonstrates an important principle of the Keystra system: Monitoring is still a risk-management action. A Watchlist classification does not mean the supplier is irrelevant, rather it means the current risk level does not justify the same level of intervention as the higher risk groups. These suppliers should therefore remain visible, with their risk indicators reassessed as new procurement, supplier-performance and dependency information becomes available.

Low Current Risk

The remaining 24 suppliers are classified as Low Current Risk. These suppliers currently have the lowest levels of combined risk within the assessed population and therefore receive a Routine intervention priority. They are:

- 1. S049 — FlowTech Engineering Services Ltd. | 31.5 | Downstream Propagation**
- 2. S066 — ChemCore Process Materials Ltd. | 30.4 | Downstream Propagation**
- 3. S041 — Apex Valve Technologies Ltd. | 29.6 | Business Unit Reach**
- 4. S020 — Electra Grid Solutions Ltd. | 29.0 | Business Unit Reach**
- 5. S012 — Summit Rotating Technologies Ltd. | 28.5 | Critical Asset Exposure**
- 6. S050 — Westline Piping Solutions Ltd. | 28.3 | Downstream Propagation**
- 7. S072 — SafeGuard Industrial Supplies Ltd. | 28.1 | Business Unit Reach**
- 8. S028 — Precision Control Technologies Ltd. | 28.0 | Downstream Propagation**
- 9. S056 — ProServe Industrial Supplies Ltd. | 27.7 | Downstream Propagation**
- 10. S005 — Northstar Mechanical Solutions Ltd. | 27.4 | Downstream Propagation**
- 11. S075 — WorkSafe Industrial Ltd. | 26.3 | Downstream Propagation**
- 12. S073 — ProShield Safety Systems Ltd. | 26.1 | Business Unit Reach**
- 13. S052 — HighPoint Valve Engineering Ltd. | 24.0 | Business Unit Reach**
- 14. S048 — Streamline Valve & Piping Ltd. | 22.8 | Downstream Propagation**

- 15. S074 — Guardian PPE Solutions Ltd. | 21.9 | Downstream Propagation
- 16. S078 — SafetyFirst Industrial Supplies Ltd. | 21.8 | Business Unit Reach
- 17. S051 — BlueRiver Flow Systems Ltd. | 20.0 | Downstream Propagation
- 18. S077 — SureSafe Protective Solutions Ltd. | 19.7 | Downstream Propagation
- 19. S002 — Meridian Mechanical Technologies Ltd. | 19.1 | Downstream Propagation
- 20. S067 — Apex Industrial Chemicals Ltd. | 15.4 | Business Unit Reach
- 21. S070 — Advanced Lubricants & Chemicals Ltd. | 14.8 | Business Unit Reach
- 22. S068 — PrimeLube Energy Services Ltd. | 12.1 | Downstream Propagation
- 23. S045 — PrimeFlow Industrial Solutions Ltd. | 0.0 | Supplier Failure Probability
- 24. S085 — WindSun Energy Systems Ltd. | 0.0 | Supplier Failure Probability

Routine classification does not mean these suppliers should be ignored permanently. It simply reflects their current risk position based on the information available to the system.

This distinction becomes particularly important for S045 and S085, both of which received a score of 0.0. Their position should not be interpreted as proof that they cannot experience future problems. Rather, their current classification indicates that they do not presently generate a stronger risk signal within the final scoring framework.

The Final Risk Picture

When the complete 90-supplier population is considered together, the structure becomes clear. There is:

- 1 Critical Early Warning supplier requiring immediate attention.
- 10 High Early Warning suppliers requiring active management.
- 21 Emerging Risk suppliers requiring planned investigation and mitigation.
- 34 Watchlist suppliers requiring enhanced monitoring.
- 24 Low Current Risk suppliers requiring routine monitoring.
- The total outstanding supplier exposure is approximately \$4.05 billion, with an exposure-weighted risk score of 57.97/100.

The dominant risk driver across the population is Downstream Propagation, reinforcing one of the central findings of the supply-chain analysis: supplier risk does not stop at the supplier.

A supplier may support a material.

That material may support an asset.

That asset may support a business unit.

And that business unit may depend on the continued availability of that supplier for safe and reliable operations.

The final risk classification therefore reflects the interconnected nature of the Keystra supply chain, and brings us to the end of this solution.